

# What Do Sequence Models Actually Use?

## Dissecting Order-Sensitivity and Permutation-Invariance with a Non-Linguistic Token Stream

**Anthony Engelmann**

Heinrich Heine University Düsseldorf

University ID: 3160882 • *Python for NLP* - Final Project

July 23, 2026

**Code:** <https://github.com/anthonyengelmann/sequence-order-4nlp>

### Abstract

Sequence models fundamentally assume that order carries information. However, because natural language deeply entangles sequence order with semantic content, isolating the true contribution of order alone remains remarkably difficult. Here, we use single-cell gene-expression profiles as a controlled, non-linguistic testbed, treating cells as order-free sets of (gene, value) tokens with artificially assigned sequence positions. We evaluate how the standard NLP architecture ladder (FNN, LSTM, Transformer) handles this set-structured signal. While a standard diagnostic task saturates all models ( $\approx 0.99$  AUROC), a severe rare-class stress test (spike-in dilution to prevalence  $\pi = 10\% \rightarrow 0.1\%$ ) paired with a within-seed analysis reveals stark architectural gaps. Once patient-split variance is canceled, the order-agnostic feedforward model proves to be the best calibrated (ECE superior by 0.017–0.020, statistically significant), while the complex sequence models remain virtually indistinguishable from one another. Imposed sequence orderings yield null results, and granting the Transformer positional encodings actively harms rare-class detection. We conclude that sequence order is redundant *once an explicit value channel already carries the signal*, and that sequence models are not merely indifferent to the redundant ordering but are actively harmed by it, making permutation-invariance the correct inductive bias for set-structured inputs.

## 1 Introduction

Modern sequence models, whether built on recurrence or attention, underpin tasks from machine translation [1] to task-oriented dialogue [2], and rest on a fundamental assumption: token *order* carries critical information. In natural language, however, order is hopelessly entangled with semantic content. You cannot shuffle the words in a sentence without destroying its meaning, making it impossible to cleanly measure how much a model relies on positional information versus the actual token identities. To untangle this, we use single-cell gene-expression profiles as a deliberately non-linguistic testbed. Biologically, a cell is simply a *set* of expressed genes with no intrinsic sequence. Any order we impose, whether by expression rank, at random, or by an importance proxy, is purely an experimental *choice*. Treating each token as an explicit (identity, value) pair allows us to ablate sequence order directly and ask a question that text simply forbids: is positional information entirely redundant when an explicit value channel exists?

## Research questions.

- Q1** Does a sequential inductive bias actually help when the underlying signal is inherently a *set*? (Evaluating feedforward vs. LSTM vs. Transformer architectures).
- Q2** If we *impose* an arbitrary sequence order, does an inherently order-sensitive model find a way to exploit it? (Evaluated via a strict ordering ablation and paired within-seed analysis).

**Contributions.** Our primary contributions are threefold: **(i)** we introduce a controlled, syntax-free testbed to isolate the effect of sequential inductive biases; **(ii)** we design a paired within-seed evaluation protocol that cuts through dominating patient-split noise to reveal true architectural effects; and **(iii)** we demonstrate empirically that for set-structured data, adding complex sequential machinery provides no measurable discrimination benefit while actively harming model calibration. Ultimately, imposed order is not exploitable.

## 2 Background and Related Work

### 2.1 The Cell as a Weighted Bag-of-Words

We encode each cell as a padded, masked sequence of gene-ID tokens, where every token also carries a continuous expression value. This mirrors standard NLP preprocessing closely enough to be worth spelling out: the cell is a *document*, its expressed genes are *tokens*, and the expression value of a token behaves like a *term frequency*. The dense input to our feedforward baseline is therefore a frequency-weighted bag-of-words [3], and selecting highly variable genes plays a role loosely analogous to inverse document frequency [4]. That makes our main question: if the signal lies in which tokens appear and how strongly, rather than in how they are arranged, does order add anything a weighted bag does not already capture?

### 2.2 Order-Sensitivity and Permutation-Invariance

Our three models differ mainly in how much they care about order. The feedforward baseline is a bag of tokens and ignores order by construction. The LSTM reads the tokens as a sequence and predicts from its final hidden state. The Transformer with masked mean-pooling is permutation-invariant until a positional encoding is added. The usual approach to unordered data is to build invariance into the architecture by hand, as Deep Sets [5] does. We come at it from the other side and ask whether ordinary order-sensitive models *behave* as if they were invariant once the signal is a set. The claim is behavioural rather than mechanistic: we measure whether held-out performance changes when the tokens are re-ordered, not whether the internal representations are order-free. The positional-encoding toggle is our one exact check, because without it a masked-mean encoder is permutation-invariant up to floating-point precision.

### 2.3 Single-Cell Foundation Models

The two best-known single-cell foundation models answer our question in opposite ways. Geneformer [6] uses *rank-value* encoding and orders genes by expression, so position carries signal by design. scGPT [7] instead bins the expression values and uses an attention mask that imposes no gene order at all. Both are large pretrained systems, so their disagreement about order is entangled with differences in scale and training corpus and has never been tested on its own. Our from-scratch ladder is built to settle it: Q1 asks whether an order-sensitive model beats an order-agnostic one on identical tokens, and Q2 asks, in Geneformer’s own terms, whether an imposed expression-rank order is exploitable at all.

### 3 Data

#### 3.1 Source and Cohort

We derive our dataset from the Single-cell Pediatric Cancer Atlas (ScPCA), specifically project SCPCP0000008 [8], which is released under a CC-BY 4.0 licence. After filtering for the 2000 most highly variable genes, the final cohort encompasses 84 patients and 480,584 individual cells (Table 1). The primary task is binary classification, distinguishing malignant *blast* cells from normal cells. The dataset is naturally skewed, with an overall blast prevalence of 87%.

#### 3.2 Preprocessing

To construct our non-linguistic token stream, we apply standard log-normalization to the raw gene counts. Each cell is then strictly reduced to a sequence of its top  $k = 256$  expressed genes. Crucially, each token encapsulates both the gene identity and an explicit, continuous expression-value channel, padded to a fixed length using standard masking techniques.

#### 3.3 Patient-Level Splits

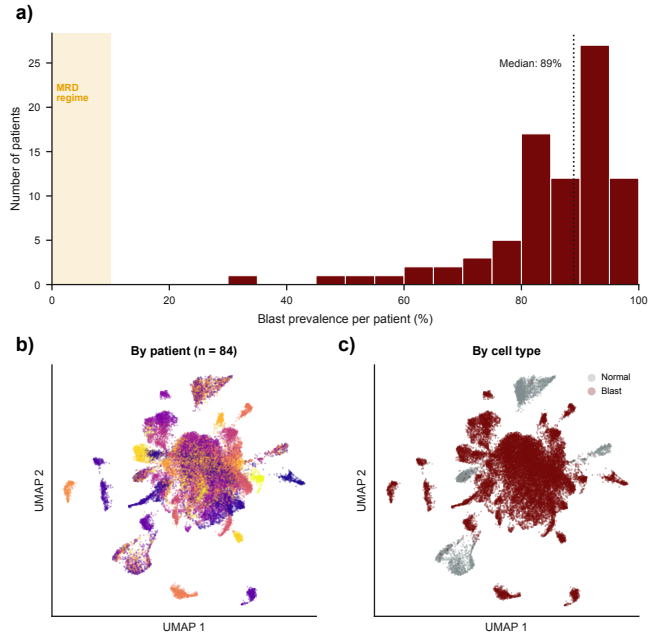
To prevent data leakage and rigorously evaluate model calibration under patient shift, all data splits (70/15/15) are performed strictly at the patient level using `GroupShuffleSplit`. This strict separation guarantees that no single patient ever appears in more than one subset.

#### 3.4 Biases and Limitations

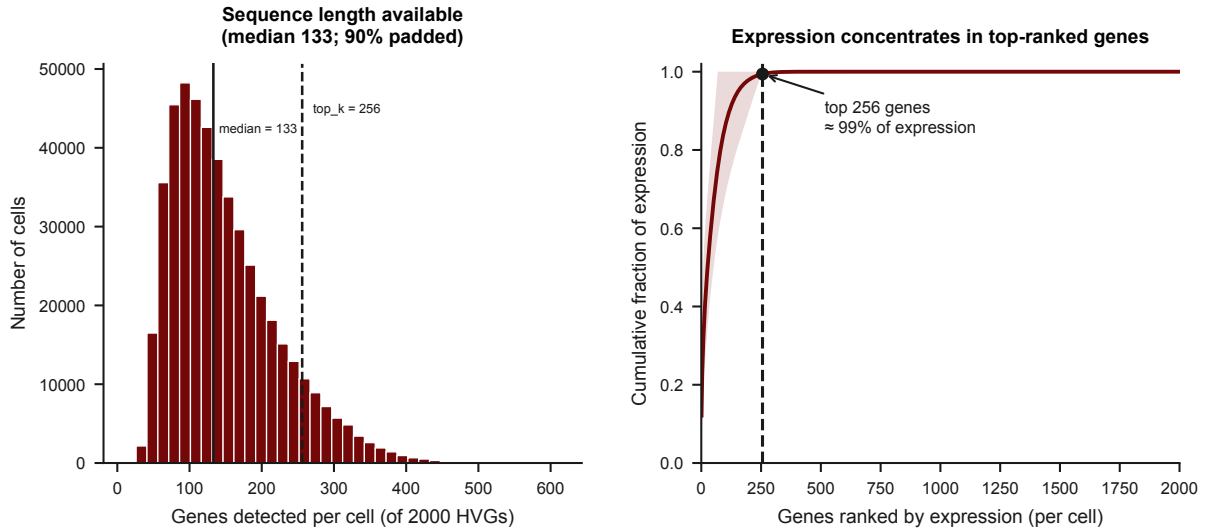
We explicitly use this pediatric B-cell leukemia cohort as an isolated, structural instrument to test architectural inductive biases. The dataset features severe natural class imbalance and targets a highly specific clinical demographic. Consequently, this study makes absolutely no clinical or diagnostic claims.

**Table 1.** Cohort characteristics (ScPCA SCPCP0000008).

| Characteristic                            | Value              |
|-------------------------------------------|--------------------|
| Patients – $n$                            | 84                 |
| Total cells – $n$                         | 480,584            |
| Blast cells – $n$ (%)                     | 417,600 (87%)      |
| Normal cells – $n$ (%)                    | 62,984 (13%)       |
| Cells per patient – median [range]        | 5,533 [973–14,659] |
| Genes per cell (2000 HVGs) – $n$          | 2,000              |
| Blast prevalence (%) – median [range]     | 89 [32–99]         |
| Age at diagnosis (years) – median [range] | 7.4 [0.6–18.8]     |
| <b>Sex – <math>n</math> (%)</b>           |                    |
| Male                                      | 50 (60%)           |
| Female                                    | 34 (40%)           |
| <b>Diagnosis – <math>n</math> (%)</b>     |                    |
| B-cell ALL                                | 84 (100%)          |
| <b>Tissue – <math>n</math> (%)</b>        |                    |
| Bone marrow                               | 81 (96%)           |
| Peripheral blood                          | 3 (4%)             |



**Figure 1. The Clinical Evaluation Gap and Patient-Driven Variance.** (a) Blast-prevalence histogram (median 89%) and the rare-class regime. (b, c) UMAPs show cells cluster by patient identity, not phenotype.



**Figure 2. Empirical justification for  $K = 256$ .** The median cell expresses 133 of 2,000 HVGs (left); the top 256 ranked genes capture  $\approx 99\%$  of a cell’s expression (right), meaning a length-256 window loses negligible signal while bounding attention cost.

## 4 Methodology and Experimental Setup

### 4.1 Representation: a cell as a sequence of gene tokens

We cast each cell as a short “sentence”: the cell is a *document*, its expressed genes are *tokens*, and the 2,000 highly variable genes (HVGs) plus one padding symbol form a vocabulary of  $|V| = 2,001$ . Unlike text, a cell has no intrinsic order, it is an unordered *bag* of  $(\text{gene-ID}, \text{expression-value})$  pairs, so every token carries both a discrete identity and an explicit continuous value channel. Each cell is encoded end-to-end (Fig. 3): its 2,000-HVG vector is (i) sorted by descending expression, (ii) truncated to the top  $K = 256$  genes, yielding paired token-ID and value vectors, (iii) padded to length 256 with  $\text{PAD} = 2000$  and a boolean mask that zeroes padded positions in attention, and (iv) mapped through a learnable embedding (gene identity,  $d=128$ ) and a linear projection (expression value,  $d=128$ ), summed elementwise into a  $[256, 128]$  tensor for the LSTM and Transformer. The FNN baseline instead consumes the dense 2,000-vector directly. The cap  $K = 256$  is empirically grounded (Fig. 2): the median cell expresses only 133 of 2,000 HVGs, and the 256 highest-ranked genes already capture  $\approx 99\%$  of a cell’s total expression. A length-256 window therefore spans the full expressed repertoire of  $\sim 90\%$  of cells while bounding the  $O(K^2)$  cost of self-attention; the resulting  $\sim 48\%$  padded tail (median  $133/256$ ) is rendered inert by masking.

### 4.2 Ordering as the controlled variable

Because the order is *imposed*, we treat it as the experimental knob and vary it while holding the token set fixed: **rank** (default, descending magnitude), **random**, **ascending** (recency probe, the highest-expression gene lands in the LSTM’s final-hidden slot), **importance-first/-last**, and a sinusoidal positional-encoding (PE) toggle [9] for the Transformer.

This gives a two-sided test of whether order is exploitable signal for a bag-of-genes: the order-sensitive LSTM asks whether imposed order *helps*, while the otherwise permutation-invariant Transformer asks whether *granting* position helps (the no-PE Transformer doubling as a float-exact invariance check).

### 4.3 Model Hierarchy

We evaluate a structural ladder of increasing order-sensitivity:

1. **Feedforward Bag-of-Tokens:** Order-agnostic baseline.
2. **LSTM:** Order-sensitive, read out from final hidden state.
3. **Transformer Encoder:** Permutation-invariant unless positional encoding is toggled.

### 4.4 Training Protocol & Hyperparameters

All models are optimized via Adam ( $\text{lr} = 10^{-3}$ , weight decay =  $10^{-5}$ ) with batch size 256 and BCEWithLogits loss. Model selection and early stopping (patience 5) use validation loss only; every reported metric is computed once on the held-out test split. We execute 10 independent random seeds (1–10), where each seed strictly fixes the patient-level split.

### 4.5 Rare-Class Stress Test

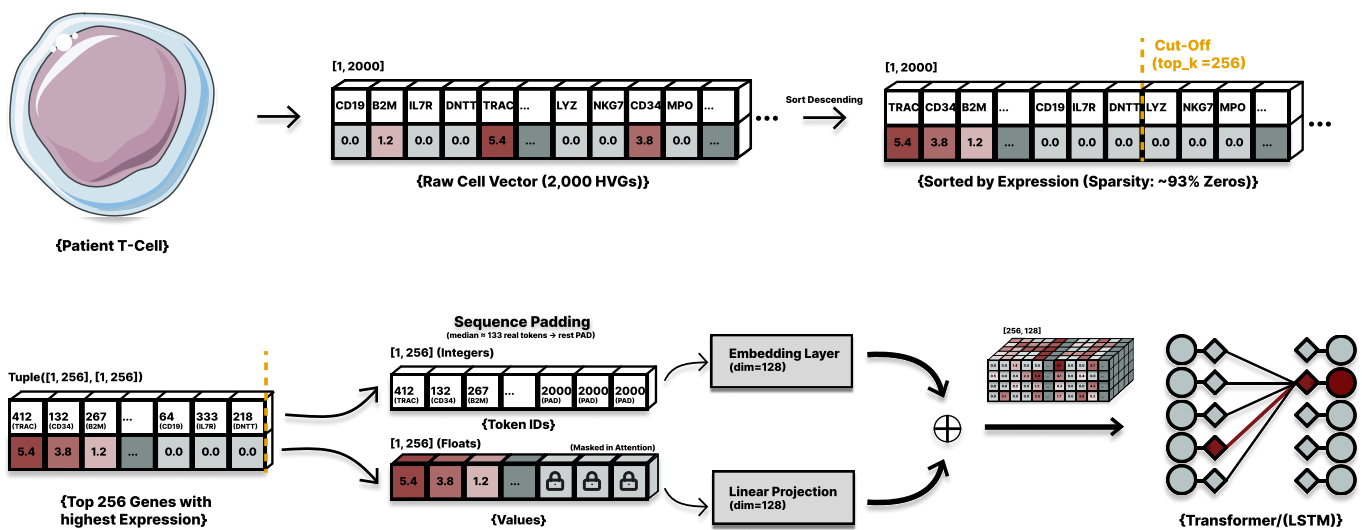
Because standard balanced classification rapidly saturates all architectures ( $\approx 0.99$  AUROC), we introduce a severe stress test by spike-in diluting the positive rate to  $\pi \in \{10\%, 1\%, 0.1\%\}$ . We report AUPRC, Sensitivity@FPR (sensitivity at a fixed false-positive rate of 0.1%, i.e. one false positive per 1,000 normal cells), and Expected Calibration Error (ECE) [10].

### 4.6 Paired Within-Seed Analysis

Patient variation heavily dominates marginal performance. Because a random seed enforces the exact same held-out patient cohort for every model, we evaluate models on a paired basis. Subtracting scores on the same seed cancels patient-split variance. We report paired difference ( $\Delta \pm \text{SE}$ ), significance ratio ( $\sigma = \Delta/\text{SE}$ ), and sign-consistency.

### 4.7 Environment & Reproducibility

Executed in PyTorch [11] on Apple-silicon MPS. Splitting, I/O, and configuration rely on scikit-learn [12], scanpy/AnnData [13], and Hydra [14].



**Figure 3. From a cell to a token sequence.** A cell (a set of expressed genes) is encoded as (gene-ID, value) tokens; the imposed ordering is the controlled variable of Q2.

## 5 Results

### 5.1 Q1: The architecture ladder on a set-structured signal

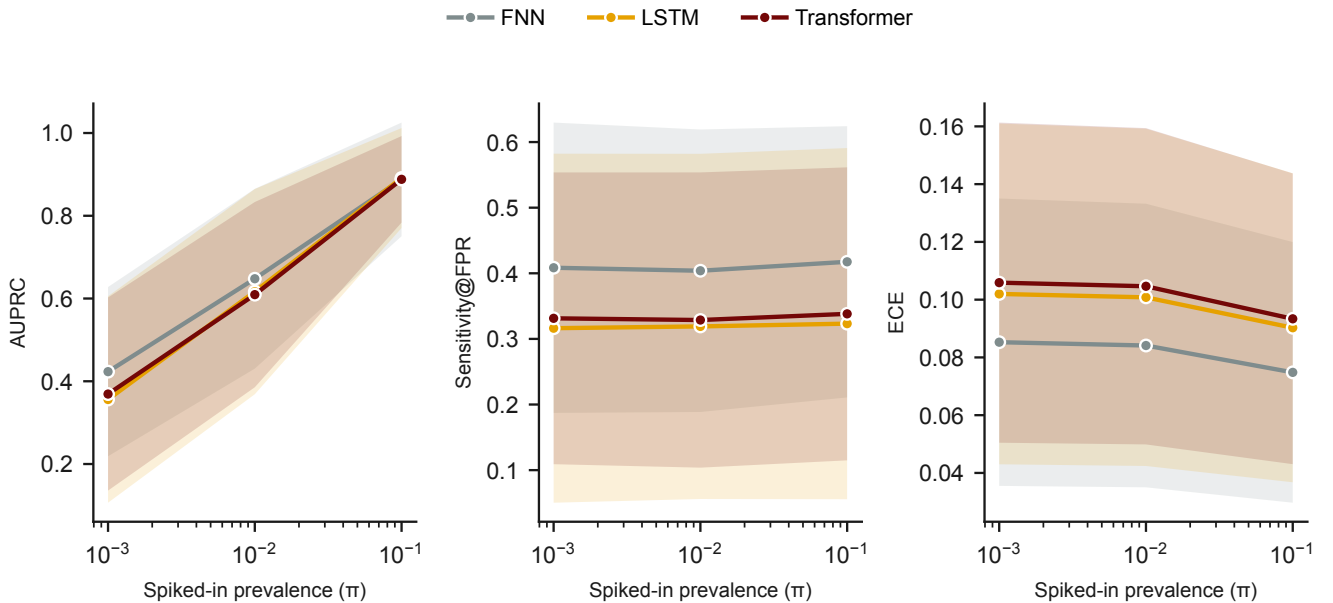
**The balanced task as a saturation control.** We first evaluate the models on the standard balanced classification task, which effectively serves as a saturation control. As detailed in Table 2, all three architectures rapidly reach near-ceiling test discrimination. When we isolate the paired within-seed differences, the performance gap vanishes completely, yielding  $|\Delta\text{AUROC}| < 0.001$  for every single contrast. Because this highly saturated environment cannot meaningfully rank the underlying inductive biases, it directly necessitates our shift to the severe rare-class stress test.

**Table 2. Balanced task — saturation control.** Test-set metrics (mean  $\pm$  SD over 10 seeds). All three architectures reach near-ceiling discrimination, so the balanced task cannot rank the inductive biases.

|              | AUROC               | AUPRC               | ECE                 |
|--------------|---------------------|---------------------|---------------------|
| Architecture |                     |                     |                     |
| FNN          | $0.9872 \pm 0.0181$ | $0.9974 \pm 0.0034$ | $0.0142 \pm 0.0100$ |
| LSTM         | $0.9878 \pm 0.0148$ | $0.9974 \pm 0.0027$ | $0.0120 \pm 0.0059$ |
| Transformer  | $0.9867 \pm 0.0140$ | $0.9972 \pm 0.0026$ | $0.0146 \pm 0.0078$ |

**Dilution separates the ladder in favor of simplicity.** Under the pressure of the rare-class stress test (Figure 4), the architectural ladder clearly separates. Strikingly, it separates in favor of the simplest approach. The order-agnostic feedforward model proves to be consistently the best calibrated. It achieves a paired  $\Delta\text{ECE}$  of  $-0.017$  against the LSTM (winning in 9 out of 10 seeds,  $\sigma \approx -2.6$ ) and  $-0.020$  against the Transformer ( $\sigma \approx -2.4$ ). In stark contrast, the two complex sequence models perform virtually identically to one another, showing no meaningful calibration difference ( $\text{ECE } \Delta \approx -0.004$ ,  $\sigma \approx -0.5$ ).

**MRD limit-of-detection — Where the ladder separates**

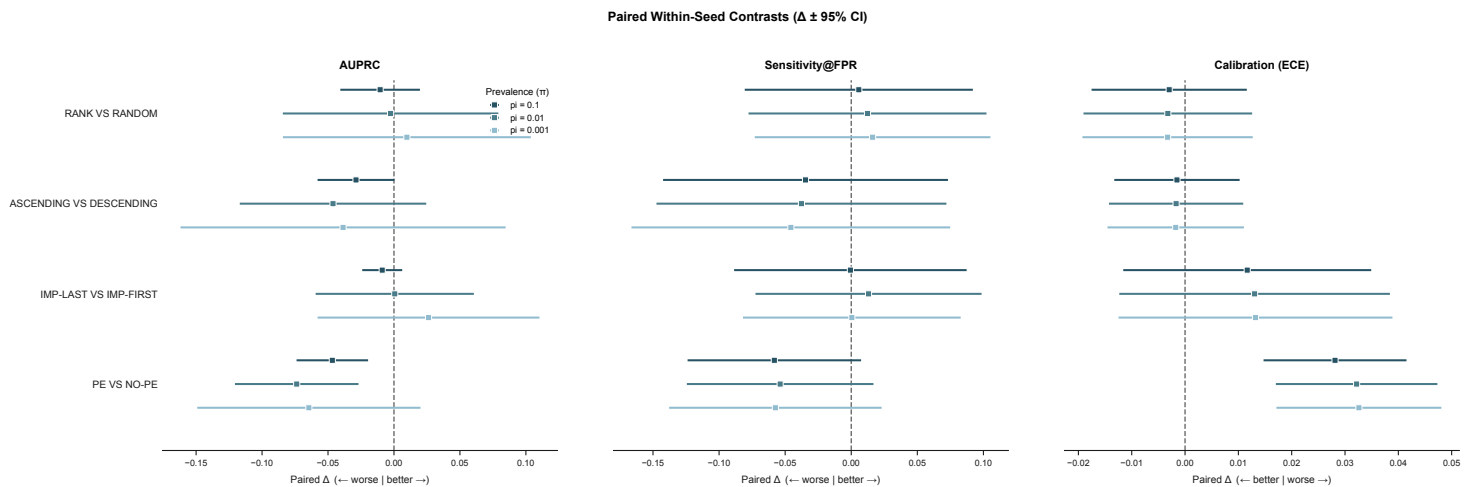


**Figure 4. Q1 — Architecture ladder under the rare-class stress test.** Rare-class metrics vs. prevalence  $\pi$ . The feedforward bag is the best-calibrated model at every  $\pi$ .

## 5.2 Q2: Is imposed order exploitable?

To determine whether an order-sensitive architecture can actually exploit an artificially imposed sequence, we analyze the paired within-seed differences on our rare-class readout. We focus here on the headline prevalence of  $\pi = 1\%$ , with the full spectrum of  $\pi$  levels detailed in Figure 5. The ablation reveals a clear consensus across every tested spatial configuration:

- **Rank versus random ordering** yields a null result ( $\text{AUPRC } \Delta = -0.003 \pm 0.042$ ,  $\sigma \approx -0.1$ ). This confirms that the LSTM extracts no detectable actionable signal from an imposed rank order compared to completely shuffled data.
- **Ascending versus descending ordering** shows only a weak, statistically insignificant anti-recency trend ( $\sigma \approx -1.3$ ).
- **Importance-last versus importance-first ordering** is similarly null ( $\Delta = +0.001 \pm 0.031$ ,  $\sigma \approx 0$ ). Even though the LSTM generates its prediction from its final hidden state, deliberately placing the most important tokens at this terminal recency slot provides no measurable benefit. The explicit expression-value channel already fully encodes the necessary signal.
- **The impact of positional encoding (PE)** produces the only statistically significant effect in this suite, and it is strictly negative. Granting the permutation-invariant Transformer access to sinusoidal spatial coordinates actively degrades its rare-class detection capabilities ( $\text{AUPRC } \sigma \approx -3.1$ , with the PE-enabled model performing worse in 9 out of 10 seeds). Furthermore, it severely harms model calibration ( $\text{ECE } \sigma \approx +4.1$ ).



**Figure 5. Q2 — ordering ablation, paired within-seed across prevalence  $\pi$ .** Every imposed ordering is null (intervals straddle 0); only positional encoding moves the model, and it moves it the wrong way.

## 6 Discussion, Error Analysis, and Limitations

### 6.1 Structure hurts when the data is a set

Do sequence models actually need order for a cell? No. On set-structured tokens, recurrence and self-attention give no edge over a bag-of-tokens and, worse, hurt calibration. It is not an information bottleneck: the median cell expresses only  $\sim 133$  genes, below our  $K = 256$  cap, so every model sees the same data. The gap is pure inductive bias: a permutation-invariant model fits the unordered reality of a cell and stays best calibrated, while the sequence models waste capacity on syntactic structure that isn't there.



## 6.2 Error analysis: The patient is the variance

Marginally the architectures look tied, but the cross-seed variance is driven by *which* patients are held out, not the architecture. Our weakest split, Seed 8, drops to mean AUROC 0.946 (vs. the 0.997 ceiling elsewhere), and one of its 13 held-out patients, SJPHALL021, causes almost all of it: leave-one-patient-out recovers 4.8 AUROC points (the next worst adds only 0.24) and it alone accounts for 41% of the seed’s errors. It is unusually normal-heavy (79% blast vs. the 90% test average), so the blast-prior models over-predict on it, which is exactly why we use a paired within-seed design.

## 6.3 Back to language

The claim is ultimately about NLP, and the cell is the control condition text cannot provide: because language never lets us switch order off, position’s contribution can only be measured on a stream where order is ours to remove. Read back, our result predicts that wherever token *identity* and *magnitude* already carry the signal (bag-of-words classification, keyword retrieval, tabular-to-text pipelines), positional machinery should be similarly inert or harmful, as the sinusoidal encoding is here. The non-linguistic testbed is thus not a departure from NLP but the instrument that makes an otherwise untestable claim about order measurable.

## 6.4 Computational cost and limitations

A few boundaries are worth pushing in future work:

- **Computational cost:** all runs used one Apple-silicon GPU (PyTorch MPS); the LSTM was the most expensive arm (its recurrence cannot be parallelised), so early stopping and capping at 500 cells per patient kept the 10-seed sweep tractable.
- **Sample size:** only ~13 held-out patients per split, on a single pediatric B-ALL dataset; more seeds and cohorts would add statistical power.
- **Training from scratch:** we trained from scratch to isolate architectural bias; large-scale pretraining (single-cell foundation models) could change how these architectures behave.
- **Clinical realism:** the rare-class stress test is a controlled, artificial dilution, a tool to probe limits, not a clinical diagnostic setting.
- **Shared, untuned configuration:** we fixed the optimiser, capacity, and schedule across the ladder rather than tuning each model separately; a per-model search could narrow the gap, though the balanced-task parity ( $\approx 0.99$  AUROC for all three) makes a reversal unlikely.

## 7 Conclusion

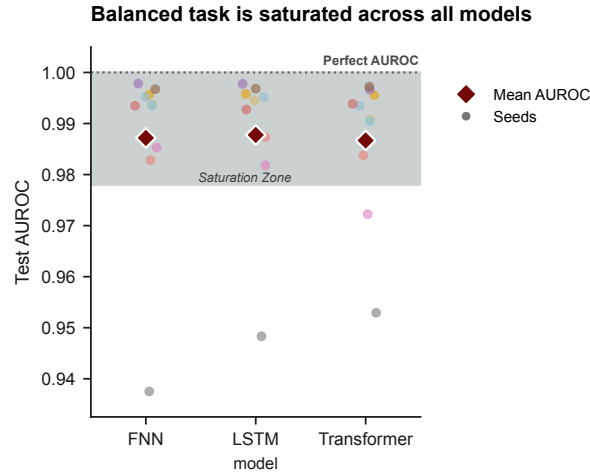
Using an order-free token stream as a controlled instrument, we find that the sequential and attentional biases central to NLP give *no benefit* over a permutation-invariant bag when the signal is a set, and that the simplest model is best calibrated under imbalance (Q1). Imposed order is not exploitable, and granting the Transformer positional structure actively hurts rare-class detection (Q2). The sharper reading is conditional: order is redundant *once an explicit value channel already carries the signal*, and models are not merely indifferent to the redundant ordering but are harmed by it. Both effects surface only under a rare-class stress test and only through paired within-seed analysis, because patient-split variance dominates the marginal comparison. Future work: remove the value channel to test position once magnitude is hidden, and repeat with pretrained encoders.



## References

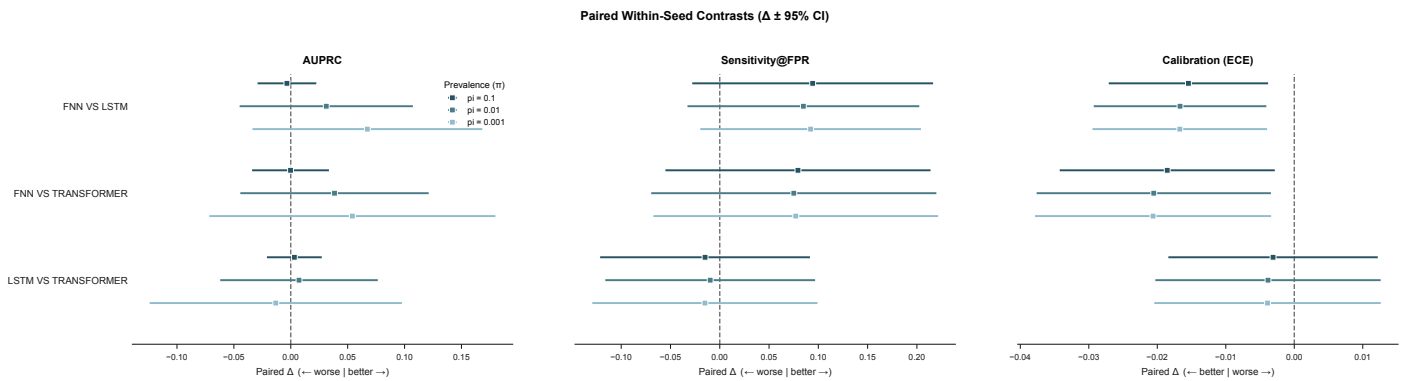
- [1] D. Bahdanau, K. Cho, and Y. Bengio, “Neural machine translation by jointly learning to align and translate,” in *Proc. ICLR*, 2015.
- [2] S. Feng, N. Lubis, C. Geishauser, H.-C. Lin, M. Heck, C. van Niekerk, and M. Gašić, “EmoWOZ: A large-scale corpus and labelling scheme for emotion recognition in task-oriented dialogue systems,” in *Proc. LREC*, 2022, pp. 4096–4113.
- [3] G. Salton and C. Buckley, “Term-weighting approaches in automatic text retrieval,” *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.
- [4] K. Spärck Jones, “A statistical interpretation of term specificity and its application in retrieval,” *Journal of Documentation*, vol. 28, no. 1, pp. 11–21, 1972.
- [5] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. Salakhutdinov, and A. J. Smola, “Deep sets,” in *Proc. NeurIPS*, 2017.
- [6] C. V. Theodoris, L. Xiao, A. Chopra, M. D. Chaffin *et al.*, “Transfer learning enables predictions in network biology,” *Nature*, vol. 618, pp. 616–624, 2023.
- [7] H. Cui, C. Wang, H. Maan, K. Pang, F. Luo, N. Duan, and B. Wang, “scGPT: Toward building a foundation model for single-cell multi-omics using generative AI,” *Nature Methods*, vol. 21, pp. 1470–1480, 2024.
- [8] Alex’s Lemonade Stand Foundation Childhood Cancer Data Lab, “The single-cell pediatric cancer atlas (ScPCA), project SCPCP000008,” <https://scpca.alexlemonade.org>, 2024, licence CC-BY 4.0.
- [9] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Proc. NeurIPS*, 2017.
- [10] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. ICML*, 2017.
- [11] A. Paszke, S. Gross, F. Massa, A. Lerer *et al.*, “PyTorch: An imperative style, high-performance deep learning library,” in *Proc. NeurIPS*, 2019.
- [12] F. Pedregosa, G. Varoquaux, A. Gramfort *et al.*, “Scikit-learn: Machine learning in Python,” *JMLR*, vol. 12, pp. 2825–2830, 2011.
- [13] F. A. Wolf, P. Angerer, and F. J. Theis, “SCANPY: Large-scale single-cell gene expression data analysis,” *Genome Biology*, vol. 19, p. 15, 2018.
- [14] O. Yadan, “Hydra – a framework for elegantly configuring complex applications,” GitHub, 2019, <https://github.com/facebookresearch/hydra>.

## A Balanced Ladder Plot



**Figure 6. Balanced task — saturation control.** All three architectures reach near-ceiling discrimination, so the balanced task cannot separate the inductive biases.

## B Dilution Forest Plot



**Figure 7. Paired within-seed contrasts ( $\Delta \pm 95\%$  CI).** LSTM and Transformer are statistically equivalent.

## C Full rare-class results (all prevalences)

**Table 3. Rare-class stress test — full results** (mean  $\pm$  SD over 10 seeds) at every spike-in prevalence  $\pi$ . AUPRC and Sensitivity@FPR measure rare-cell detection; ECE measures calibration.

| Architecture | $\pi$ (Prevalence) | AUPRC               | Sensitivity@FPR     | ECE                 |
|--------------|--------------------|---------------------|---------------------|---------------------|
| FNN          | 0.100000           | 0.8879 $\pm$ 0.0707 | 0.4175 $\pm$ 0.2659 | 0.0748 $\pm$ 0.0038 |
|              | 0.010000           | 0.6478 $\pm$ 0.1854 | 0.4039 $\pm$ 0.2655 | 0.0841 $\pm$ 0.0031 |
|              | 0.001000           | 0.4232 $\pm$ 0.2563 | 0.4085 $\pm$ 0.2859 | 0.0853 $\pm$ 0.0023 |
| LSTM         | 0.100000           | 0.8913 $\pm$ 0.0439 | 0.3233 $\pm$ 0.1222 | 0.0903 $\pm$ 0.0038 |
|              | 0.010000           | 0.6166 $\pm$ 0.1021 | 0.3190 $\pm$ 0.1305 | 0.1008 $\pm$ 0.0032 |
|              | 0.001000           | 0.3559 $\pm$ 0.1438 | 0.3163 $\pm$ 0.1609 | 0.1020 $\pm$ 0.0022 |
| Transformer  | 0.100000           | 0.8881 $\pm$ 0.0566 | 0.3382 $\pm$ 0.1618 | 0.0934 $\pm$ 0.0041 |
|              | 0.010000           | 0.6094 $\pm$ 0.1325 | 0.3288 $\pm$ 0.1669 | 0.1046 $\pm$ 0.0035 |
|              | 0.001000           | 0.3691 $\pm$ 0.1695 | 0.3314 $\pm$ 0.1930 | 0.1059 $\pm$ 0.0024 |

## D Hyperparameters

**Table 4.** Model and training configuration. Shared settings apply to all three architectures unless noted; **ordering** and positional encoding are the ablation levers. The models are comparable in size (0.5–0.65 M trainable parameters), so the calibration gap reflects inductive bias, not capacity.

| Setting                                                       | Value                                                                  |
|---------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Shared — optimisation &amp; training</b>                   |                                                                        |
| Optimizer                                                     | Adam (lr $10^{-3}$ , weight decay $10^{-5}$ )                          |
| Loss                                                          | BCEWithLogits                                                          |
| Batch size                                                    | 256                                                                    |
| Max epochs / early stopping                                   | 50 / patience 5 on val. loss ( $\Delta_{\min} = 10^{-4}$ )             |
| Seeds                                                         | 1–10 (seed also fixes the patient split)                               |
| Patient-level split (GroupShuffleSplit)                       | 70 / 15 / 15 % (train / val / test)                                    |
| Calibration bins (ECE)                                        | 10                                                                     |
| Hardware                                                      | single Apple-silicon GPU (PyTorch MPS)                                 |
| <b>Shared — token representation (LSTM &amp; Transformer)</b> |                                                                        |
| Vocabulary                                                    | 2000 HVGs + 1 PAD = 2001                                               |
| Sequence length (top- $k$ )                                   | 256, rank-ordered, PAD = 2000 (attention-masked)                       |
| Token channels                                                | gene-ID embedding $\oplus$ expression value ( $d = 128$ )              |
| <b>FNN — baseline (dense input)</b>                           |                                                                        |
| Architecture / dropout                                        | MLP 2000 $\rightarrow$ 256 $\rightarrow$ 1, ReLU / 0.2 (0.51 M params) |
| <b>LSTM — order-sensitive (token input)</b>                   |                                                                        |
| Embedding / hidden dim                                        | 128 / 256                                                              |
| Layers / dropout                                              | 1, unidirectional, final-hidden readout / 0.2 (0.65 M params)          |
| <b>Transformer — self-attention (token input)</b>             |                                                                        |
| $d_{\text{model}}$ / heads                                    | 128 / 4                                                                |
| Encoder layers / FFN dim                                      | 2 / 512 (GELU), post-LayerNorm residual                                |
| Attention / pooling                                           | softmax (dense) / masked mean (0.65 M params)                          |
| Positional encoding                                           | none (default)   sinusoidal (ablation)                                 |

## Declaration of AI Usage

I used an AI assistant (Claude Opus 4.8) throughout this project, mainly for brainstorming and for non-learning “plumbing”. Concretely, it assisted with: framing the research questions and the analysis strategy; the scaffolding and configs, testing and debugging, the data pipeline (loading, preprocessing, and exploratory data analysis); plotting and figure generation; result aggregation and the paired-analysis tooling; parts of the evaluation metrics and the rare-class evaluation; the experiment-launcher, demo, and configuration scripts; and the drafting, structuring, and language editing of this report, including its tables and captions. The parts that are central to the course, the model architectures (FNN, LSTM, Transformer) and the training procedure, I implemented myself, and I ran and interpreted all experiments. I re-ran or independently checked every reported number against my own outputs, tested and where needed corrected any AI-assisted code, and I take full responsibility for the correctness of the entire submission. No AI tool is cited as an academic source, and no confidential or personal data were shared with the AI service (the dataset is public, released under CC-BY 4.0).